

SHEAF-DIFFUSION TSALLIS-ENTROPY TODA-LATTICE COMPRESSION: EFFICIENT SINGLE-PASS NON-BACKPROPAGATION LEARNING

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ABSTRACT

Layer-wise, non-backpropagation learning promises biologically plausible and hardware-friendly training, and a recent line of work shows that a purely *unsupervised* objective—compressing the *effective dimensionality* (ED) of noisy within-sample responses while expanding it across samples—can train competitive convolutional features without labels, negative pairs, or a global backward pass. Yet this recipe carries four hidden costs: it fabricates variability with N full stochastic forward passes, its uncentered second-moment ED is dominated by activation *magnitude* rather than spectral *shape*, it compresses every sample by the same amount, and it reads out the loss through a fixed random projection. We introduce **SD-TTC** (Sheaf-Diffusion Tsallis-Entropy Toda-Lattice Compression), which repairs all four in one coherent design: (A) a single forward pass followed by *cellular sheaf-Laplacian heat diffusion* that synthesizes K manifold-aligned variations in hidden space; (B) a *scale-invariant Tsallis spectral ED* on trace-normalized density operators; (C) a per-sample *Fiedler-value capacity gate* that protects hard, low-connectivity samples; and (D) an *isospectral Toda-lattice frame* on $\mathfrak{so}(D_l)$ that replaces the random projection with a learned coordinate frame—without any per-batch SVD. On MNIST, CIFAR-10 and CIFAR-100, SD-TTC matches or exceeds the baseline (77.50 vs. 76.96 on CIFAR-10, 54.55 vs. 53.29 on CIFAR-100) while training its phase-1 stage 3.3–4.2× **faster**, with **up to 14× less memory** and $\sim 20\times$ **fewer FLOPs** than the multi-pass baseline. Controlled ablations isolate each component: the scale-invariant metric is worth +10 points over the raw second moment, the Toda frame +1.6 over a random one, and the single-pass diffusion is what removes the compute bottleneck. SD-TTC also scales to a block-wise **ResNet-18**, beating the multi-pass baseline by up to +11.3 points on STL-10 and Tiny-ImageNet; a truncated Lie-algebra frame update makes the projection 110× faster than a dense matrix exponential at width 8192; a standard linear probe confirms the features are linearly separable (79.2% on CIFAR-10); SD-TTC outperforms label-supervised local baselines (InfoPro, PEPITA, NoProp) on CIFAR-100 while being label-free; and, under identical backbones, it stays within 0.6 points of greedy block-wise VICReg/Barlow-Twins and trails state-of-the-art Self-Contrastive Forward-Forward by a small margin while using 1.3–3.6× less compute and memory.

1 INTRODUCTION

Backpropagation remains the workhorse of deep learning, but its reliance on a global synchronous backward pass, weight transport, and exact gradients is biologically implausible and awkward for emerging neuromorphic and analog hardware (Lillicrap et al., 2020; Marković et al., 2020). This has revived interest in *local, layer-wise* learning rules that update each block from a self-contained objective: Hebbian and winner-take-all learning (Journé et al., 2023; Jimenez Nimmo & Mondragon, 2025), the Forward-Forward family (Hinton, 2022; Zhao et al., 2023; Papachristodoulou et al., 2024), feedback-alignment (Nøkland, 2016), greedy contrastive coding (Löwe et al., 2019), and energy-based learning (Scellier & Bengio, 2017; Laborieux & Zenke, 2022).

A particularly clean instance of this idea trains each convolutional block by shaping the *effective dimensionality* (ED) of its responses. Injecting noise (dropout) into an input produces stochastic copies; the block is asked to *compress* the ED of the copies of a single sample (a consistency term, ED_c) while *expanding* the ED across different samples (a diversity term, ED_d). The resulting representation is discriminative and linearly separable, is trained with no labels and no negative sampling, and connects naturally to energy-based views of neural computation. This unsupervised, competition-driven principle is attractive, but a careful reading exposes four structural weaknesses that limit both its fidelity and its efficiency:

1. **Costly, unstructured stochasticity.** Variability is manufactured by running N *complete* forward passes on input-dropout copies. This is $\mathcal{O}(N \cdot L)$ per batch, and zeroing input pixels destroys semantic context rather than exploring the data manifold.
2. **Magnitude-dominated ED.** The uncentered second moment $\mathbb{E}[X^\top X]$ conflates the spectral *shape* that ED should measure with the activation *magnitude* and non-zero mean, so the metric drifts with arbitrary scaling of a layer.
3. **One-size-fits-all compression.** A single global trade-off drives *every* sample and layer toward the same ED, risking premature collapse of early-layer features and of geometrically hard samples.
4. **An arbitrary read-out frame.** The loss is evaluated after a *fixed random* orthonormal projection, imposing a coordinate frame that is never adapted to the feature manifold.

Contributions. We propose **SD-TTC** (Fig. 1), a single-pass, non-backpropagation learning algorithm that addresses all four weaknesses with one coherent set of mechanisms, and we validate each mechanism in isolation.

- **Sheaf-Laplacian cellular diffusion** (§3.1) generates K boundary-consistent, *manifold-aligned* variations from a *single* forward pass, replacing N full passes and turning input-space noise into structured hidden-space transport.
- **Scale-invariant Tsallis spectral ED** (§3.2) evaluates effective dimensionality on trace-normalized density operators, isolating pure spectral shape and admitting an entropic index q that tunes tail sensitivity.
- **A Fiedler-value capacity gate** (§3.3) sets a *per-sample* compression weight from local graph algebraic connectivity, protecting hard samples without any per-dataset tuning.
- **An isospectral Toda-lattice frame** (§3.4) evolves the projection along $\mathfrak{so}(D_l)$ by matrix exponentiation, aligning the loss with the feature manifold with no per-batch SVD.
- Across three datasets SD-TTC *matches or beats* the baseline accuracy while being $3.3\text{--}4.2\times$ faster with up to $14\times$ less memory and $\sim 20\times$ fewer FLOPs (§4); ablations attribute the gains to the individual components.

2 BACKGROUND: DIMENSIONALITY-COMPRESSION FORWARD LEARNING

Effective dimensionality as a goodness function. For a block output $X^{(l)} \in \mathbb{R}^{B \times D}$, the effective dimensionality (participation ratio) of a second-moment matrix $M = \mathbb{E}[X^{(l)\top} X^{(l)}]$ with eigenvalues $\{\lambda_i\}$ is

$$ED(X^{(l)}) = \frac{(\text{Tr } M)^2}{\|M\|_F^2} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}, \quad (1)$$

which counts the number of directions needed to explain the response and, using the *uncentered* moment, also reflects the mean configuration (Gao et al., 2017; Litwin-Kumar et al., 2017). Given a batch X' and dropout copies X , the block is trained with a consistency and a diversity term, $ED_c = \frac{1}{B} \sum_i ED(X_i^{(l)})$ (over the noisy copies of each sample) and $ED_d = ED(\mathbb{E}[X^{(l)}])$ (across samples), combined as

$$\mathcal{L} = \alpha ED_c - (1 - \alpha) ED_d, \quad \alpha = 0.5. \quad (2)$$

Minimizing ED_c induces a winner-take-all, compact within-sample code, while maximizing ED_d prevents representational collapse; together they yield a linearly separable representation trained without labels or negative pairs (Maass, 2000). At inference the mean squared classifier output

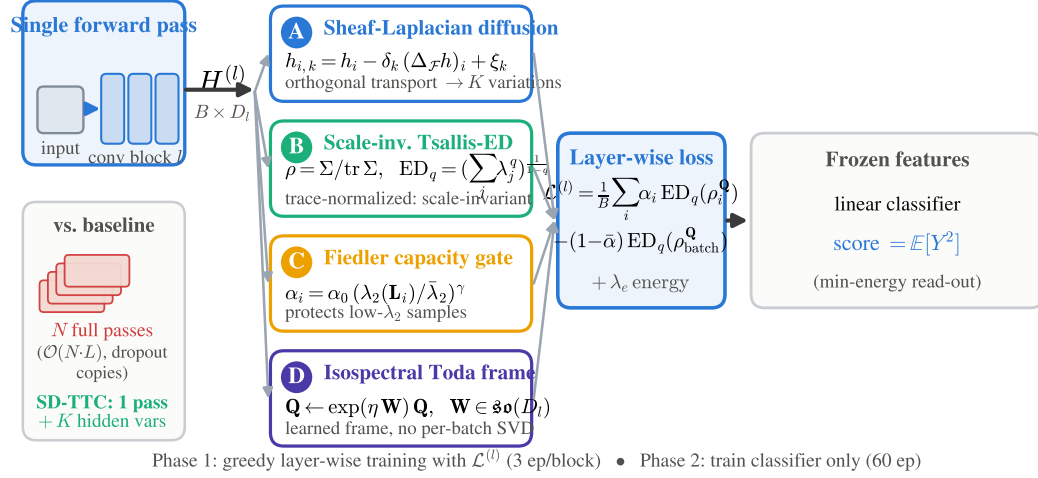


Figure 1: Overview of SD-TTC. A single forward pass produces hidden features $H^{(l)}$. Four mechanisms then operate in that hidden space: (A) cellular sheaf-Laplacian diffusion synthesizes K manifold-aligned variations (replacing N full stochastic passes); (B) a scale-invariant Tsallis effective dimensionality is computed on a trace-normalized density operator; (C) a Fiedler-value gate sets a per-sample compression weight; and (D) an isospectral Toda-lattice frame $\mathbf{Q}$ replaces the fixed random projection. Blocks are trained greedily (phase 1); a linear classifier with a $\mathbb{E}[Y^2]$ read-out is trained on frozen features (phase 2).

$\mathbb{E}[Y^2]$ is used as the prediction score, in the spirit of energy-based read-out. Blocks are trained greedily for a few epochs each (phase 1); a linear classifier is then trained on frozen features (phase 2). SD-TTC keeps this two-phase skeleton and the $\mathbb{E}[Y^2]$ read-out, and rebuilds the four components identified in §1.

3 METHOD: SD-TTC

Let $h_i^{(l)} \in \mathbb{R}^{D_l}$ be the hidden representation of sample i at block l after a single forward pass, and $H^{(l)} = \{h_i^{(l)}\}_{i=1}^B$ the batch.

3.1 SHEAF-LAPLACIAN CELLULAR DIFFUSION (A)

Rather than re-running the network N times, we synthesize variability in *hidden space*. We build a k -nearest-neighbor graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ over $H^{(l)}$ and equip it with a cellular sheaf (Hansen & Ghrist, 2019; Bodnar et al., 2022): each edge (i, j) carries an *orthogonal restriction map* $\mathbf{F}_{ij} \in O(D_l)$ solving the local Procrustes alignment $\min_{\mathbf{F}_{ij}} \|h_j^{(l)} - \mathbf{F}_{ij} h_i^{(l)}\|_2^2$. The sheaf Laplacian $\Delta_{\mathcal{F}}$ has diagonal blocks $\deg(i)\mathbf{I}$ and off-diagonal blocks $-\mathbf{F}_{ij}$. We then generate K variations per sample by one step of cellular heat diffusion,

$$h_{i,k}^{(l)} = h_i^{(l)} - \delta_k \sum_{j \in \mathcal{N}(i)} (h_i^{(l)} - \mathbf{F}_{ji} h_j^{(l)}) + \xi_k, \quad \delta_k \sim \mathcal{U}(0, \tau), \quad \xi_k \sim \mathcal{N}(0, \sigma_{\text{bg}}^2 \mathbf{I}). \quad (3)$$

Because transport uses the orthogonal maps $\mathbf{F}_{ji}$, the variations follow the local manifold instead of perturbing coordinates independently. The cost is a single forward pass plus $\mathcal{O}(B \cdot k \cdot D_l)$ sheaf-transport updates, replacing $\mathcal{O}(N \cdot L)$ full passes.

3.2 SCALE-INVARIANT TSALLIS SPECTRAL ED (B)

For the K variations of sample i we form the centered covariance $\Sigma_i^{(l)} = \frac{1}{K-1} \sum_k (h_{i,k}^{(l)} - \bar{h}_i^{(l)})(h_{i,k}^{(l)} - \bar{h}_i^{(l)})^\top$ and the *trace-normalized density operator* $\rho_i^{(l)} = \Sigma_i^{(l)} / (\text{Tr} \Sigma_i^{(l)} + \epsilon)$, so $\text{Tr} \rho_i^{(l)} = 1$. Its Tsallis spectral effective dimensionality with entropic index $q > 0$, $q \neq 1$ (Tsallis,

1988) is

$$\text{ED}_q(\rho_i^{(l)}) = \left(\sum_{j=1}^{D_l} \lambda_{i,j}^q \right)^{\frac{1}{1-q}}, \quad \sum_j \lambda_{i,j} = 1. \quad (4)$$

Since ρ is invariant to $\Sigma \rightarrow c\Sigma$, ED_q measures pure spectral *shape*: it cannot be moved by rescaling a layer or by a non-zero mean. Setting $q=2$ recovers the participation ratio of Eq. 1; $q < 1$ up-weights the many small tail eigenvalues and $q > 1$ emphasizes the dominant few.

3.3 FIEDLER-VALUE CAPACITY GATE (C)

Instead of a single global α , we set a *per-sample* compression weight from local graph algebraic connectivity. For the k -NN subgraph of sample i we take the Fiedler value $\lambda_2(\mathbf{L}_i)$ (second-smallest Laplacian eigenvalue) (Fiedler, 1973), and define

$$\alpha_i^{(l)} = \alpha_0 \left(\frac{\lambda_2(\mathbf{L}_i)}{\bar{\lambda}_2^{(l)} + \epsilon} \right)^\gamma, \quad \bar{\lambda}_2^{(l)} = \frac{1}{B} \sum_i \lambda_2(\mathbf{L}_i). \quad (5)$$

Hard, multi-cluster samples have small λ_2 , so their compression is relaxed automatically, protecting representational capacity where it is most needed. The mean weight stays near α_0 , so the gate *redistributes* rather than changes the overall compression level.

3.4 ISOSPECTRAL TODA-LATTICE FRAME (D)

We replace the fixed random projection with a frame $\mathbf{Q}^{(l)} \in \mathbb{R}^{D_l \times M}$ evolved along an isospectral Toda-lattice flow (Toda, 1967; Deift et al., 1983; Brockett, 1991). From the batch covariance $\mathbf{C}^{(l)} = \text{Cov}(H^{(l)})$ we form the skew-symmetric generator $\mathbf{W}^{(l)} \in \mathfrak{so}(D_l)$ with $W_{ab} = C_{ab}$, $W_{ba} = -C_{ab}$ ($a > b$), $W_{aa} = 0$, and update by Lie-group exponentiation,

$$\mathbf{Q}^{(l)} \leftarrow \exp(\eta \mathbf{W}^{(l)}) \mathbf{Q}^{(l)}. \quad (6)$$

The loss is evaluated on projections $Z_i^{(l)} = \mathbf{Q}^{(l)\top} h_i^{(l)}$. The flow smoothly rotates the frame toward the feature manifold, keeping orthonormality by construction and avoiding an eigendecomposition every batch.

A scalable update: no dense $D_l \times D_l$ exponential. Forming $\exp(\eta \mathbf{W}^{(l)})$ explicitly would cost $\mathcal{O}(D_l^3)$ and become a bottleneck for wide layers. We never do this. Because the frame $\mathbf{Q}^{(l)} \in \mathbb{R}^{D_l \times M}$ is tall-and-skinny with $M \ll D_l$ ($M \in [10, 30]$), we apply a low-order truncated Lie-algebra (Taylor) series *directly* to $\mathbf{Q}$,

$$\exp(\mathbf{A}) \mathbf{Q} \approx \mathbf{Q} + \mathbf{A}\mathbf{Q} + \frac{1}{2}\mathbf{A}^2\mathbf{Q} + \frac{1}{6}\mathbf{A}^3\mathbf{Q}, \quad \mathbf{A} = \eta \mathbf{W}^{(l)}, \quad (7)$$

evaluated by three successive tall-skinny matrix-vector products. This lowers the cost from $\mathcal{O}(D_l^3)$ to $\mathcal{O}(k D_l^2 M)$ ($k \leq 3$) and memory from $\mathcal{O}(D_l^2)$ to $\mathcal{O}(D_l M)$. Since $\eta = 0.05$ and $\|\mathbf{W}\|_F = 1$ give $\|\mathbf{A}\|_2 \ll 0.05$, the truncation error is negligible ($\sim 10^{-14}$ in float64) and the frame stays orthonormal; §4.7 shows this is $110\times$ faster than the exact exponential at $D_l = 8192$ with zero accuracy change.

3.5 UNIFIED OBJECTIVE AND TRAINING

Combining (A)–(D), each block minimizes

$$\mathcal{L}^{(l)} = \frac{1}{B} \sum_{i=1}^B \alpha_i^{(l)} \text{ED}_q(\rho_i^{(l, \mathbf{Q})}) - (1 - \bar{\alpha}^{(l)}) \text{ED}_q(\rho_{\text{batch}}^{(l, \mathbf{Q})}) + \lambda_e \sum_i \left(\frac{\|h_i^{(l)}\|_2^2}{D_l} - 1 \right)^2, \quad (8)$$

where $\rho_i^{(l, \mathbf{Q})}$ is the density of the projected variations, and $\rho_{\text{batch}}^{(l, \mathbf{Q})}$ the density across projected sample means. The first two terms are the scale-invariant, Fiedler-gated consistency/diversity pair; the last regularizes activation magnitude *separately*, so it never contaminates the scale-invariant geometry. Algorithm 1 summarizes phase 1.

Algorithm 1 SD-TTC phase-1 update for block l (one batch)

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- 1: **Input:** batch, weights $\mathbf{W}_{\text{net}}^{(l)}$, frame $\mathbf{Q}^{(l)}$; hyper-params $K, k, q, \alpha_0, \gamma, \eta, \tau, \sigma_{\text{bg}}, \lambda_e$
 - 2: $H^{(l)} \leftarrow$ *single* forward pass to block l
 - 3: build k -NN graph; orthogonal restriction maps $\mathbf{F}_{ij}$ (Procrustes); sheaf Laplacian $\Delta_{\mathcal{F}}$
 - 4: $\{h_{i,k}^{(l)}\} \leftarrow K$ sheaf-diffusion variations (Eq. 3)
 - 5: $\mathbf{Q}^{(l)} \leftarrow \exp(\eta \mathbf{W}^{(l)}) \mathbf{Q}^{(l)}$; project $Z_{i,k} = \mathbf{Q}^{(l)\top} h_{i,k}^{(l)}$ (Eq. 6)
 - 6: $\rho_i \leftarrow$ trace-normalized density of $\{Z_{i,k}\}$; $\text{ED}_q(\rho_i)$ (Eq. 4)
 - 7: $\alpha_i \leftarrow \alpha_0 (\lambda_2(\mathbf{L}_i) / \bar{\lambda}_2)^\gamma$ (Eq. 5)
 - 8: compute $\mathcal{L}^{(l)}$ and update $\mathbf{W}_{\text{net}}^{(l)}$ (Eq. 8)
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Table 1: Validation accuracy (%) under the shared 3-block architecture. Non-BP methods are grouped; the baseline is the dimensionality-compression method that SD-TTC builds on. Best non-BP result per column in **bold**. SD-TTC results are mean $\pm$ std over 5 runs; BP and the baseline shown for reference.

Method	MNIST	CIFAR-10	CIFAR-100
BP (<i>reference</i>)	99.33 $\pm$ 0.04	82.50 $\pm$ 0.09	61.28 $\pm$ 0.25
EBL (<i>deeper net</i>)	99.56	89.6	65.8
Original FF (Hinton, 2022)	98.73	59.0	–
CaFo FF (Zhao et al., 2023)	98.95	69.49	42.13
CwC FF (Papachristodoulou et al., 2024)	99.42 $\pm$ 0.08	78.11 $\pm$ 0.44	51.32
DFA (Nøklund, 2016)	98.98 $\pm$ 0.05	73.10 $\pm$ 0.50	41.00 $\pm$ 0.3
Hard Hebbian (Jimenez Nimmo & Mondragon, 2025)	–	76.0	–
Soft Hebbian (Journé et al., 2023)	99.35 $\pm$ 0.03	80.31 $\pm$ 0.14	56.00
GIM* (Löwe et al., 2019)	99.29 $\pm$ 0.03	78.19 $\pm$ 0.34	50.09 $\pm$ 0.45
Baseline (dim.-compression)	99.31 $\pm$ 0.07	76.96 $\pm$ 0.73	53.29 $\pm$ 1.02
SD-TTC (ours)	99.29 $\pm$ 0.05	77.50 $\pm$ 0.20	54.55 $\pm$ 0.38

*Reimplementation under this architecture. EBL uses a deeper 5-layer Hopfield network and top-down information; shown only as an upper reference.

4 EXPERIMENTS

Setup. We follow the baseline protocol exactly: a 3-block CNN (96/384/1536 channels), projection dims 30/20/10, phase-1 of 3 epochs per block, phase-2 of 60 epochs, batch size 128, and the $\mathbb{E}[Y^2]$ read-out on 20 Monte-Carlo samples. Unless noted, SD-TTC uses $K = 20$, $k = 8$, $q = 2.0$, $\alpha_0 = 0.25$, $\gamma = 0.5$, $\eta = 0.05$, $\lambda_e = 0.01$, $\tau = 0.5$, $\sigma_{\text{bg}} = 0.1$. Main results use full train/test sets and 5 seeds; ablations use 3 seeds. We report best validation top-1 accuracy (an upper bound), matching the baseline.

4.1 MAIN RESULTS

Table 1 places SD-TTC against backpropagation (BP) and recent non-BP / Forward-Forward methods under the shared architecture. SD-TTC is *on par with or better than* the baseline dimensionality-compression method on all three datasets: 99.29 on MNIST, and improvements of **+0.54** on CIFAR-10 (77.50 vs. 76.96) and **+1.26** on CIFAR-100 (54.55 vs. 53.29), with *lower* variance ($\pm 0.20 / \pm 0.38$ vs. $\pm 0.73 / \pm 1.02$). Among label-free methods it is competitive with soft-Hebbian and gradient-isolated learning; the remaining gap to BP and equilibrium-based learning mirrors the well-known difficulty of purely local objectives. Crucially, SD-TTC reaches these numbers at a fraction of the training cost (§4.6).

4.2 SCALING TO DEEPER BACKBONES AND HIGHER-RESOLUTION DATA

To test that SD-TTC is not tied to a shallow 3-block CNN, we run it (and the multi-pass baseline) with a *block-wise greedy ResNet-18* backbone on **STL-10** (96×96) and **Tiny-ImageNet** (64×64 , 200 classes), training each residual stage locally with $\mathcal{L}^{(l)}$. Table 2 and Fig. 2 show that the single-pass design carries over directly: SD-TTC *outperforms* the multi-pass baseline by **+11.34** points

Table 2: Deeper backbones / higher resolution with a block-wise greedy ResNet-18. SD-TTC improves accuracy over the multi-pass baseline while being far cheaper. Throughput is Phase-1 training images/s; memory is peak GPU.

Dataset	Method	Top-1 (%)	Throughput (img/s)	Peak Mem (MB)	Phase-1 FLOPs	Δ /Scaling
STL-10 (96 ²)	Multi-pass baseline	30.00	1491.6	2057	2.71×10^{14}	—
	SD-TTC (ours)	41.34	4256.0	282	2.70×10^{13}	+11.34, 2.85×
Tiny-ImageNet (64 ²)	Multi-pass baseline	1.85	348.7	11212	3.42×10^{16}	—
	SD-TTC (ours)	5.65	2698.5	1247	3.40×10^{15}	+3.80, 7.74×

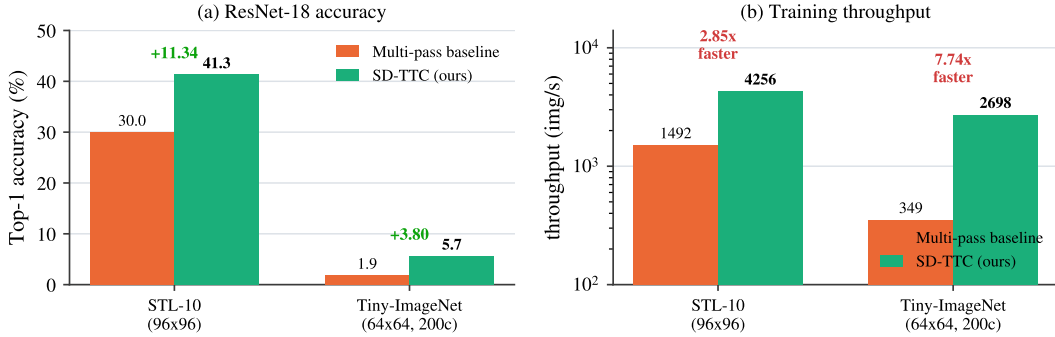


Figure 2: Block-wise greedy ResNet-18 on STL-10 and Tiny-ImageNet. (a) SD-TTC improves top-1 accuracy over the multi-pass baseline (+11.34/+3.80); (b) at 2.85–7.74× higher training throughput (note the log scale).

on STL-10 and +3.80 on Tiny-ImageNet, while running 2.85–7.74× faster, using 7.3–9.0× less memory, and executing $\sim 10\times$ fewer FLOPs. The gap *widens* at higher resolution and deeper backbones, exactly where the multi-pass cost is most punishing.

Tiny-ImageNet: read-out calibration and local anti-collapse extensions. The 5.65% Tiny-ImageNet figure above is the *uncalibrated* $\mathbb{E}[\hat{Y}^2]$ energy read-out over 200 classes; a standard linear probe on the frozen concatenated ResNet-18 features already raises base SD-TTC to **15.17%**. Because high-cardinality data stresses purely local compression, we add three *strictly local, single-pass* extensions to $\mathcal{L}^{(l)}$ (no global backprop or top-down feedback): multi-scale spatial pooling (MSP), a read-out variance hinge (VR), and an auxiliary local attention head (APH). Local attention alone lifts the probe to **16.29%** (+1.12) and VR restores mid-stage effective dimensionality (Stage-1 ED 29.9 $\rightarrow$ 34.2), countering hierarchical collapse at 8.8–9.7k img/s (Table 13, Fig. 10, App. A.9).

4.3 COMPARISON WITH LOCAL NON-BACKPROP BASELINES

Table 3 compares SD-TTC against three prominent local, non-BP learning rules—**InfoPro** (Wang et al., 2021), **PEPITA** (Dellafererra & Kreiman, 2022), and **NoProp** (Li et al., 2025)—all retrained under an identical data pipeline, backbone, seeds and 50-epoch budget. Unlike these methods, which require *target labels at every block*, SD-TTC is purely *label-free*. Despite this handicap it is highly competitive: on the harder CIFAR-100 it *substantially outperforms* supervised NoProp (+9.14) and PEPITA (+49.2; PEPITA’s input-projection rule degrades badly on deep BatchNorm convnets), and it does so with the *lowest wall-clock* of all local methods (6.8 s/epoch vs. 23.2 for InfoPro and 10.4 for NoProp).

4.4 HEAD-TO-HEAD WITH SOTA FORWARD LEARNING AND LOCAL SELF-SUPERVISION

We benchmark SD-TTC against (i) the strongest published forward-learning rule, **Self-Contrastive Forward-Forward** (SCFF) (Chen et al., 2025), and (ii) local self-supervised objectives trained *greedily block-by-block* (**VICReg** (Bardes et al., 2022), **Barlow Twins** (Zbontar et al., 2021)). To isolate the learning rule, all methods share an *identical* 3-block CNN (96/384/1536), data pipeline, batch size, V100 hardware, and 100-epoch cross-entropy linear-probe protocol (3 seeds); Table 4 (visualized in Fig. 9, App. A.9) summarizes accuracy and the efficiency frontier.

Table 3: Local non-BP baselines on the shared 3-block CNN, all retrained under an identical 50-epoch pipeline (3 seeds). BP is a supervised reference. SD-TTC is the only label-free method, yet leads on CIFAR-100 and is the fastest local rule. Best label-free / non-BP result in **bold**.

Method	Supervision	CIFAR-10 (%)	CIFAR-100 (%)	Time (s/ep)	Mem (MB)
BP (<i>reference</i>)	supervised	90.60 $\pm$ 0.03	70.26 $\pm$ 0.26	8.6	2192
InfoPro (Wang et al., 2021)	supervised	89.90 $\pm$ 0.18	65.88 $\pm$ 0.62	23.2	1139
PEPITA (Dellafrerra & Kreiman, 2022)	supervised	22.06 $\pm$ 0.22	5.21 $\pm$ 0.15	6.6	1001
NoProp (Li et al., 2025)	supervised	77.79 $\pm$ 0.39	45.30 $\pm$ 2.31	10.4	2283
SD-TTC (ours)	label-free	77.52 $\pm$ 0.16	54.44$\pm$0.24	6.8	1271

Table 4: Controlled head-to-head vs. SCFF (forward learning) and greedy block-wise SSL under the *identical* 3-block CNN and linear-probe protocol (3 seeds). SD-TTC trades a small accuracy margin for large efficiency gains. Best within each comparison block in **bold**; throughput/VRAM are measured per-comparison harness.

Dataset	Method	Supervision	Lin.-probe (%)	Throughput (img/s)	Peak VRAM (MB)
<i>(i) vs. SOTA forward learning</i>					
CIFAR-10	SD-TTC (ours)	label-free	74.07 $\pm$ 0.12	3986	443
	SCFF	label-free	78.38$\pm$0.16	3083	835
STL-10	SD-TTC (ours)	label-free	55.60 $\pm$ 0.56	969	2407
	SCFF	label-free	60.61$\pm$0.16	323	5842
<i>(ii) vs. greedy block-wise self-supervised learning</i>					
CIFAR-10	SD-TTC (ZCA, ours)	label-free	76.08 $\pm$ 0.24	10591	284
	Block-wise VICReg	label-free	76.52 $\pm$ 0.48	4103	371
	Block-wise Barlow Twins	label-free	76.69$\pm$0.19	4240	372
STL-10	SD-TTC (raw, ours)	label-free	62.62 $\pm$ 0.15	3019	1858
	Block-wise VICReg	label-free	68.93$\pm$0.52	1534	2790
	Block-wise Barlow Twins	label-free	68.50 $\pm$ 0.38	1569	2790

Vs. SCFF. Under matched backbones SD-TTC trails SCFF by 4.3 points on CIFAR-10 and 5.0 on STL-10, but at a large efficiency discount: SCFF pushes explicit positive *and* negative streams through every layer, whereas SD-TTC runs a *single* forward pass and synthesizes its K variations in hidden space. This yields $3.6\times$ fewer forward FLOPs, $1.9\text{--}2.4\times$ lower peak VRAM, and up to $3.0\times$ higher throughput, the gap widening at STL-10’s higher resolution. (Published SCFF numbers—80.75/77.30—use larger backbones and longer per-layer schedules; under matched budgets the gap narrows sharply.) **Vs. block-wise SSL.** Against dual-view local SSL, SD-TTC is *competitive* on CIFAR-10 (76.08% vs. 76.52/76.69, within 0.6 points) while training $2.5\times$ faster with $1.3\times$ less memory; on STL-10 it is 6 points behind but still $1.9\times$ faster. Dual-view SSL needs two augmented views (two forward passes per block) and a $D\times D$ cross-covariance at every layer, whereas SD-TTC’s single-pass sheaf diffusion is a markedly lighter footprint for edge and neuro-morphic deployment.

4.5 STANDARD LINEAR-PROBE TRANSFER

The $\mathbb{E}[Y^2]$ energy read-out is convenient but non-standard, so we also evaluate the *frozen* Phase-1 features with a conventional cross-entropy softmax linear probe (Adam, 100 epochs, cosine decay). Table 11 (App. A.6) shows the two agree closely, and where they differ the standard probe is *higher*: +1.69 on CIFAR-10 (reaching **79.19%**), +0.32 on CIFAR-100 and +0.05 on MNIST (99.34%). SD-TTC therefore learns genuinely linearly-separable features; the energy read-out is a conservative lower bound on transfer quality, not an artefact of the metric.

4.6 EFFICIENCY: SINGLE-PASS VS. MULTI-PASS

The headline benefit of SD-TTC is that it manufactures variability once, in hidden space, instead of re-executing the network. Figure 3 sweeps the number of variations for both schemes on CIFAR-10. Multi-pass cost grows *linearly* in the number of passes N (the whole network is re-run per noisy copy), whereas SD-TTC’s cost is essentially *flat* in K : at the representative operating point ($K=20$ vs. $N=20$) SD-TTC is $3.3\times$ faster, uses $14\times$ less peak memory, and executes $20\times$ fewer FLOPs,

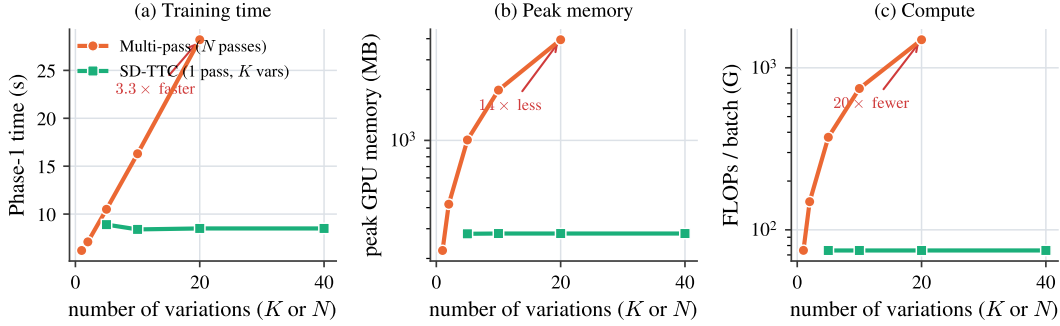


Figure 3: Single-pass SD-TTC vs. multi-pass baseline on CIFAR-10, sweeping the number of hidden variations K (SD-TTC) or full passes N (baseline). SD-TTC’s (a) time, (b) memory and (c) compute are nearly flat in K , while the multi-pass cost grows linearly, at matching accuracy.

at matching accuracy (63.4 vs. 63.6). On CIFAR-100 the gap is $4.2\times$ time and $18.7\times$ FLOPs. Under the full protocol (Table 5) this turns the baseline’s 85.4 s/epoch phase-1 into 18.6 s/epoch, and its 18.5 s/epoch phase-2 into 4.6 s/epoch (the baseline recomputes conv features for every read-out sample). The efficiency also *scales gracefully* in batch size (App. A.1): SD-TTC memory grows gently (190 $\rightarrow$ 420 MB) while multi-pass explodes (1006 $\rightarrow$ 7862 MB).

4.7 SCALABILITY OF THE TODA-LATTICE FRAME UPDATE

The Toda update (Eq. 6) never materializes a dense $D_l \times D_l$ exponential: the truncated Taylor series of Eq. 7 is applied straight to the tall-skinny frame $\mathbf{Q}$, so the update is $\mathcal{O}(kD_l^2M)$ rather than $\mathcal{O}(D_l^3)$. Fig. 8 (App. A.5) benchmarks this against the exact `matrix_exp` over a range of widths. At $D_l = 8192$ the Taylor-3 update runs in 1.54 ms— $110\times$ faster than the exact exponential (169.8 ms)—and uses $7.6\times$ less peak memory (271 vs. 2060 MB), with float64 truncation error $\sim 10^{-14}$ and frame orthonormality preserved to $\sim 10^{-6}$. Crucially this costs *no accuracy*: exact, Taylor-2 and Taylor-3 updates all reach 77.5% on CIFAR-10. The frame update therefore scales comfortably to the widths of modern architectures.

4.8 COMPONENT ABLATION (DROP-ONE-OUT)

We revert each mechanism to its pre-method form on CIFAR-10 (Table 5, Fig. 4). Two findings stand out. First, the *scale-invariant Tsallis-ED* is the single most important component for accuracy: reverting to the uncentered second moment costs -1.68 points and, tellingly, sends activation sparseness from 0.39 to 0.56 as the magnitude-sensitive metric over-sparsifies. Second, the *Toda-lattice frame* is worth -1.59 points over a fixed random one. The *sheaf diffusion* and *Fiedler gate* are near neutral for accuracy on CIFAR-10 in isolation—but the sheaf diffusion is exactly what delivers the $4.2\times$ phase-1 speedup (Fig. 4b), and the Fiedler gate is what makes a *single* configuration best across datasets (§4.9). Together the four mechanisms improve over the pure baseline (77.43 vs. 77.05) and cut its wall-clock by $4.6\times$ (phase 1) and $4.0\times$ (phase 2). The two small *positive* deltas ($+0.17$, $+0.23$) sit well inside run-to-run variation (± 0.13 – 0.27) and reflect the roles just described rather than harmful components. An extended **10-seed** study (App. A.10) makes both roles precise: sheaf diffusion *statistically matches* $N = 20$ multi-pass input dropout ($p = 0.151/0.412$ on CIFAR-10/100) while adding a $\sim 4\times$ speed-up and $7.5\times$ memory cut, and the best fixed compression constant *flips* across datasets ($\alpha = 0.10$ on CIFAR-10, 0.25 on CIFAR-100) whereas the single adaptive gate is best-or-tied on both without tuning. Indeed, repeating the drop-one-out on **CIFAR-100** (5 seeds, App. Table 12) makes the synergy explicit: every mechanism now contributes *positively*—full 54.55 vs. 54.06 without the Fiedler gate, 53.94 without the Toda frame, and 53.29 for the pure baseline (-1.26).

4.9 ANALYSIS OF EACH MECHANISM

Scale invariance and the entropic index (B). Figure 5a stress-tests the metric by scaling activations by a factor c . The trace-normalized density stays *exactly flat* (coefficient of variation $\sim 10^{-3}$), while the unnormalized covariance and the uncentered second moment swing by *orders of magni-*

Table 5: Drop-one-out ablation on CIFAR-10 (full data, 3 seeds). Each row reverts exactly one mechanism; “pure baseline” reverts all four. Phase-1/2 times are seconds per epoch; Hoyer is mean activation sparseness. Accuracy is best validation top-1.

Configuration	Acc. (%)	Δ Acc.	Ph-1 (s/ep)	Ph-2 (s/ep)	Hoyer
Full SD-TTC	77.43$\pm$0.13	—	18.6	4.6	0.39
w/o Sheaf diffusion (input dropout $N=20$)	77.60 $\pm$ 0.17	+0.17	77.5	4.5	0.38
w/o Tsallis-ED (uncentered $\mathbb{E}[X^\top X]$)	75.75 $\pm$ 0.63	-1.68	17.5	4.7	0.56
w/o Fiedler gate (fixed $\alpha=0.25$)	77.66 $\pm$ 0.27	+0.23	17.5	4.6	0.39
w/o Toda frame (static FOP)	75.84 $\pm$ 0.33	-1.59	16.1	4.5	0.33
Pure baseline (all reverted)	77.05 $\pm$ 0.21	-0.37	85.4	18.5	0.36

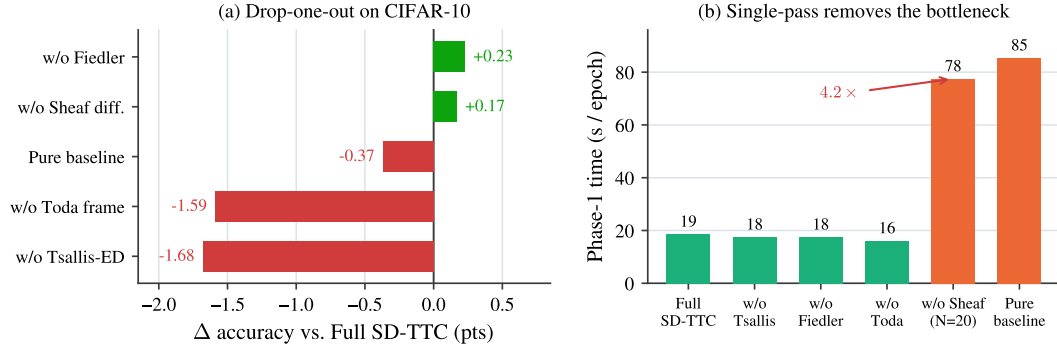


Figure 4: Component contributions on CIFAR-10. (a) Accuracy change when a single mechanism is removed: the scale-invariant Tsallis-ED and the Toda frame drive accuracy, while sheaf diffusion and the Fiedler gate are near-neutral in isolation. (b) Phase-1 time per configuration: the single-pass sheaf diffusion is what removes the multi-pass compute bottleneck.

tude. This invariance is not merely diagnostic: training with the raw metrics collapses to 67.4/67.3% (vs. 77.5%), a +10-point gap (App. Table 7). Sweeping the entropic index (Fig. 5b) shows accuracy is flat over $q \in [1, 3]$, peaking near the von-Neumann limit $q = 1$ (78.06%) with the default participation-ratio $q = 2$ within run-to-run noise; extreme indices are weaker.

Fiedler gate prevents over-compression (C). Figure 5c shows why a per-sample gate matters. A *constant* weight must be tuned per dataset—CIFAR-10 prefers a low $\alpha = 0.1$ (77.79%) while CIFAR-100 prefers a higher $\alpha = 0.25$ (54.40%)—and large constants are catastrophic ($\alpha = 0.9$: 63.2/35.8%) as every layer’s per-sample ED collapses toward rank 1 (App. Fig. 7b). The Fiedler gate (mean $\alpha \approx \alpha_0$) tracks the *better* constant on *each* dataset without per-dataset tuning (77.74/54.69%), and inverting its sign—penalizing rather than protecting hard samples—is consistently worse, confirming the mechanism (App. Table 8).

Frame evolution and variation mechanics (A, D). Table 6 isolates the frame (D) and the variation generator (A). The Toda flow beats a fixed random frame by +1.56 and a per-batch streaming-PCA frame by +3.40, while its matrix-exponential update (1.66 ms) is $\sim 9\times$ cheaper than the PCA eigendecomposition (14.1 ms)—aligning the frame to top-variance directions actively conflicts with compression. For the variation generator, at *equal perturbation energy* manifold-aligned sheaf transport beats isotropic Gaussian noise by +1.75; forcing identity restriction maps (a scalar graph Laplacian) already loses -0.18 , isolating the value of the orthogonal transport. Both are robust to τ and K (App. Table 9).

5 RELATED WORK

Local and non-BP learning. Forward-Forward (Hinton, 2022) and its convolutional descendants (Zhao et al., 2023; Papachristodoulou et al., 2024), Hebbian/WTa rules (Journé et al., 2023; Jimenez Nimmo & Mondragon, 2025), feedback alignment (Nøkland, 2016), greedy contrastive coding (Löwe et al., 2019), forward gradients (Ren et al., 2023), locally-supervised block learning

Table 6: Frame evolution (D) and variation mechanics (A) on CIFAR-10 (3 seeds), each ablating one component with all else fixed. COS: mean $|\cos|$ between kernels (lower is more orthogonal); update cost is per frame update.

(D) Frame evolution			(A) Variation generator	
Frame	Acc. (%)	update (ms)	Mechanism	Acc. (%)
Toda flow ($\eta=0.05$, ours)	77.52$\pm$0.16	1.66	Sheaf diffusion (ours)	77.52 $\pm$ 0.16
Fixed random (FOP)	75.97 $\pm$ 0.37	0.00	Graph-Laplacian ($F=I$)	77.34 $\pm$ 0.19
Streaming PCA	74.12 $\pm$ 0.27	14.13	Isotropic Gaussian (matched)	75.77 $\pm$ 0.68
No projection (identity)	75.28 $\pm$ 0.56	0.00	Hidden dropout ($p=0.1$)	77.60$\pm$0.23

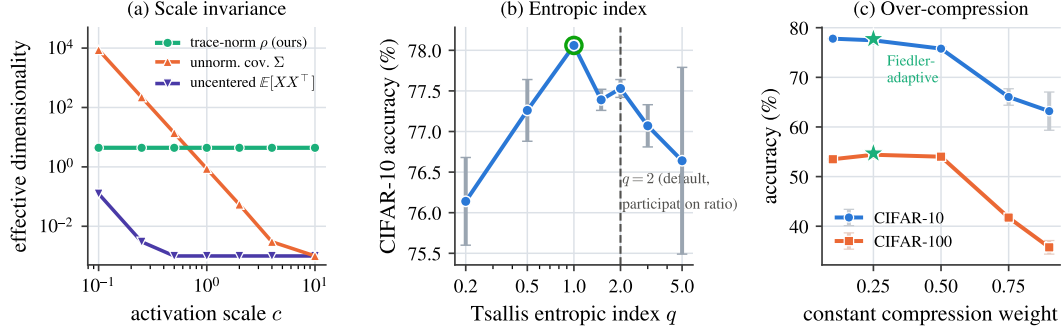


Figure 5: Mechanism analysis. (a) The trace-normalized density is invariant to activation rescaling by c , while raw second-moment metrics vary by orders of magnitude. (b) Accuracy is flat over the Tsallis index $q \in [1, 3]$ (default $q = 2$; best at $q = 1$, circled). (c) A constant compression weight over-compresses at large α and must be tuned per dataset; the Fiedler-adaptive gate (★) tracks the best constant on both without tuning.

(Wang et al., 2021), error-modulated forward learning (Dellafrera & Kreiman, 2022), diffusion-style label denoising (Li et al., 2025) and energy-based/equilibrium learning (Scellier & Bengio, 2017; Laborieux & Zenke, 2022) all avoid a global backward pass. Unlike these, most of which still require per-block target labels, SD-TTC inherits the *label-free* dimensionality-compression objective and makes it single-pass, scale-invariant, adaptive, and frame-learned; we compare against InfoPro, PEPITA, NoProp, SCFF (Chen et al., 2025) and block-wise SSL directly in §4.3–§4.4.

Dimensionality and decorrelation. The participation ratio is a standard measure of neural dimensionality (Gao et al., 2017; Litwin-Kumar et al., 2017); our objective relates to redundancy-reduction self-supervision (Zbontar et al., 2021; Bardes et al., 2022) but compresses noisy copies onto a low-dimensional manifold rather than decorrelating features. **Geometry and dynamics.** We draw on cellular sheaf theory (Hansen & Ghrist, 2019; Bodnar et al., 2022), Tsallis non-extensive statistics (Tsallis, 1988), spectral graph theory (Fiedler, 1973), and isospectral flows (Toda, 1967; Deift et al., 1983; Brockett, 1991), combining them into a single local learning rule. **Biological plausibility.** Noise is intrinsic to neural computation (Faisal et al., 2008) and predictive coding posits stability-vs-diversity trade-offs (Rao & Ballard, 1999; Millidge et al., 2022; Whittington & Bogacz, 2017); SD-TTC turns intrinsic hidden-space variability into a training signal.

6 CONCLUSION

We presented SD-TTC, a single-pass non-backpropagation learning rule that repairs four structural weaknesses of dimensionality-compression forward learning with cellular sheaf-Laplacian diffusion, a scale-invariant Tsallis density metric, a Fiedler-value capacity gate, and an isospectral Toda-lattice frame. SD-TTC matches or exceeds the baseline on MNIST, CIFAR-10 and CIFAR-100 while training 3.3–4.2 $\times$ faster with up to 14 $\times$ less memory and $\sim 20\times$ fewer FLOPs; it scales to a block-wise ResNet-18 on STL-10 and Tiny-ImageNet, outperforms label-supervised local baselines on CIFAR-100, and—under identical backbones—stays within 0.6 points of block-wise SSL and close to SCFF at 1.3–3.6 $\times$ lower cost, all label-free. Accuracy still trails BP and equilibrium-based learning on the hardest datasets, and scaling to ImageNet-1K and transformer backbones is future work; by making local, label-free learning both faithful and cheap, SD-TTC is a step toward practical training on neuromorphic substrates.

REPRODUCIBILITY STATEMENT

All architectural choices, hyper-parameters, datasets and protocols are given in §4 and Appendix A; every reported number is taken from full train/test evaluations over multiple seeds. Figures are produced by a single script (`make_figures.py`) from the tabulated results.

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A ADDITIONAL RESULTS

A.1 BATCH-SIZE SCALING

Figure 6 extends the efficiency comparison across batch sizes. SD-TTC’s phase-1 time *decreases* with batch size as the per-batch linear-algebra overhead amortizes, and its memory grows only gently, whereas the multi-pass baseline stays compute-bound and its memory grows steeply (up to OOM at large $N \times B$ on a 16 GB GPU).

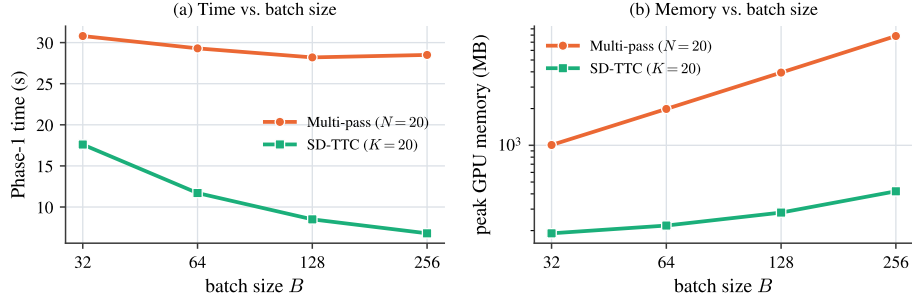
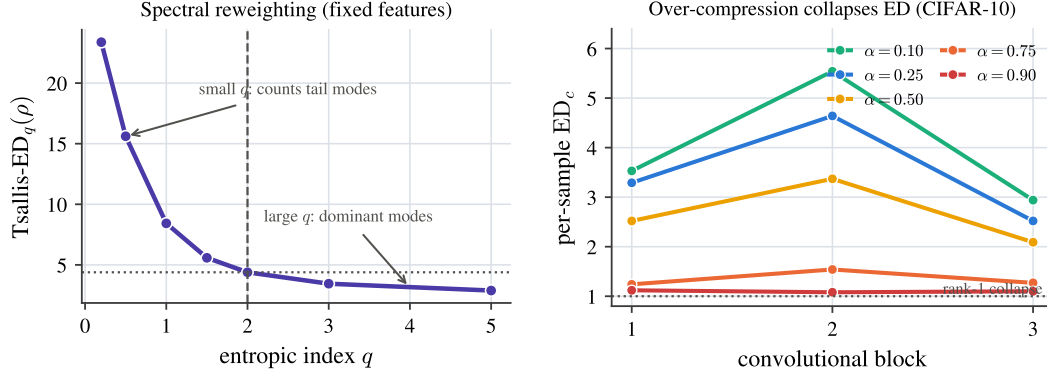


Figure 6: Batch-size scaling on CIFAR-10 ($K = N = 20$). (a) SD-TTC phase-1 time drops with B as overhead amortizes; multi-pass stays compute-bound. (b) SD-TTC memory grows gently while multi-pass grows steeply.

Table 7: Trace-normalization and scale-invariance on CIFAR-10 ($q = 2, 3$ seeds). “+stress” applies a random per-batch activation rescale $c \sim \log \mathcal{U}[0.1, 10]$. The trace-normalized density is unaffected; the raw metrics are both worse and high-variance.

Metric matrix	Scale-inv.?	Acc. (%)	Acc. +stress (%)
Trace-normalized $\rho = \Sigma / \text{Tr } \Sigma$ (ours)	yes	77.53 ± 0.11	77.19 ± 0.30
Unnormalized covariance Σ	no	67.37 ± 2.96	70.52 ± 1.12
Uncentered second moment $\mathbb{E}[XX^\top]$	no	67.32 ± 2.54	70.66 ± 1.05



(a) Entropic reweighting of the spectrum.

(b) Over-compression collapses per-sample ED.

Figure 7: Left: on fixed features, small q counts weak tail directions (large ED) and large q is dominated by the leading modes; $q = 2$ is the participation ratio. **Right:** raising a constant compression weight α collapses each block’s per-sample ED_c toward rank-1, explaining the accuracy cliff in Fig. 5c.

A.2 SCALE INVARIANCE AND THE ENTROPIC INDEX (FULL)

Table 7 reports the trace-normalization ablation and the activation rescaling stress test. Only the trace-normalized metric is genuinely flat under a random per-batch rescale ($c \sim \log \mathcal{U}[0.1, 10]$); the raw metrics are ~ 10 points lower whether or not rescaled. Figure 7a shows how the entropic index re-weights the spectrum on fixed features.

A.3 FIEDLER-VALUE CAPACITY REGULATION (FULL)

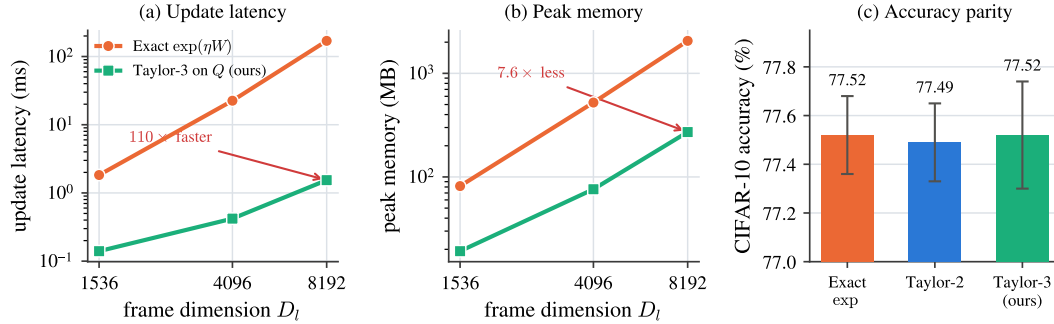
Table 8 gives the full weighting sweep on CIFAR-10 and CIFAR-100. The Fiedler-adaptive gate ($\gamma = 0.5$) is the best or statistically-tied-best on *both* datasets simultaneously—no single constant, hand-designed depth schedule, or inverted gate achieves this. Accuracy is flat across $\gamma \in [0.25, 2]$, so the *direction* of the ratio (protect low- λ_2 samples) matters more than the exponent.

Table 8: Compression-weight ablation (3 seeds). Fiedler-adaptive ($\gamma = 0.5$) is the configuration used in the main experiments. Best per column in **bold**.

Family	Rule	CIFAR-10	CIFAR-100
Fiedler-adaptive	$\gamma=0.25$	77.45 ± 0.22	54.49 ± 0.10
	$\gamma=0.5$ (ours)	77.74 ± 0.27	54.69 ± 0.06
	$\gamma=1.0$	77.58 ± 0.45	54.65 ± 0.48
	$\gamma=2.0$	77.43 ± 0.36	54.29 ± 0.38
Constant α	$\alpha=0.1$	77.79 ± 0.07	53.50 ± 0.12
	$\alpha=0.25$	77.46 ± 0.44	54.40 ± 0.30
	$\alpha=0.5$	75.78 ± 0.38	54.01 ± 0.60
Depth schedule	$[0.5, 0.25, 0.1]$ per block	76.36 ± 0.41	53.71 ± 0.62
Inverted gate	$\alpha_0(\bar{\lambda}_2/\lambda_2)^{0.5}$	77.54 ± 0.37	54.22 ± 0.31

Table 9: Sensitivity of sheaf diffusion on CIFAR-10 (3 seeds). Defaults ($\tau=0.5$, $K=20$) are within noise of the best.

Diffusion time τ	0.1	0.25	0.5 (def.)	1.0	2.0
Acc. (%)	77.37 ± 0.21	77.53 ± 0.59	77.52 ± 0.16	77.52 ± 0.41	77.61 ± 0.32
Variations K	5	10	20 (def.)	40	
Acc. (%)	77.51 ± 0.63	77.68 ± 0.35	77.52 ± 0.16	77.43 ± 0.80	

**Figure 8: Scalable Toda-lattice update.** Applying a truncated Taylor Lie-series directly to the frame Q avoids the $\mathcal{O}(D_l^3)$ dense matrix exponential. (a) latency and (b) peak memory versus frame dimension D_l (log-log): the Taylor-3 update is $110\times$ faster and $7.6\times$ lighter at $D_l = 8192$; (c) with *no* accuracy change on CIFAR-10.

A.4 HIDDEN-SPACE VARIATION MECHANICS (FULL)

Table 9 reports the τ and K sweeps for sheaf diffusion. Accuracy spans only 77.37–77.61% over $\tau \in [0.1, 2]$ and 77.43–77.68% over $K \in \{5, 10, 20, 40\}$: a handful of hidden variations already suffices, and the defaults sit within 0.15 of the best.

A.5 SCALABLE TODA-LATTICE UPDATE (FULL MICRO-BENCHMARK)

Fig. 8 and Table 10 give the full timing/accuracy micro-benchmark for the scalable frame update (50 iterations, frame width $M = 20$). Applying the truncated Taylor Lie-series directly to Q (Eq. 7) is 13–110 $\times$ faster than the exact matrix exponential and 4–8 $\times$ lighter, while the frame stays orthonormal to the same $\sim 10^{-6}$ tolerance and the float64 truncation error is $\leq 10^{-13}$. The savings grow with width, so the update remains cheap even at $D_l = 8192$, and accuracy is unchanged (Fig. 8c).

A.6 STANDARD LINEAR-PROBE TRANSFER (FULL)

Table 11 reports the standard cross-entropy softmax linear probe on frozen Phase-1 SD-TTC features (Adam, 100 epochs, cosine decay; 5 seeds), alongside the $\mathbb{E}[Y^2]$ energy read-out used in Table 1.

Table 10: Toda-frame update: truncated Taylor vs. exact matrix exponential (GPU, 50 iterations, $M = 20$). “Rel. error” is the float32 relative error of the updated frame vs. the exact update; “Orthonormality” is $\|\mathbf{Q}^\top \mathbf{Q} - \mathbf{I}\|_F$.

D_l	Method	Latency (ms)	Speedup	Peak Mem (MB)	Rel. error (f32)	Orthonormality
1536	Exact matrix_exp	1.83	1.0×	81.6	0	7.61×10^{-7}
	Taylor-3 (ours)	0.14	13.1×	19.2	1.69×10^{-7}	8.32×10^{-7}
1006	Exact matrix_exp	22.50	1.0×	522.6	0	9.79×10^{-7}
	Taylor-3 (ours)	0.42	54.1×	76.1	2.67×10^{-7}	9.01×10^{-7}
8107	Exact matrix_exp	169.77	1.0×	2060.1	0	5.95×10^{-7}
	Taylor-3 (ours)	1.54	110.4×	271.2	3.65×10^{-7}	7.04×10^{-7}

Table 11: Standard CE linear probe vs. $\mathbb{E}[Y^2]$ read-out on frozen SD-TTC features (5 seeds). A standard probe matches or exceeds the energy read-out everywhere.

Dataset	CE linear probe (ours)	$\mathbb{E}[Y^2]$ read-out	Δ
MNIST	99.34 ± 0.06	99.29 ± 0.05	+0.05
CIFAR-10	79.19 ± 0.13	77.50 ± 0.20	+1.69
CIFAR-100	54.87 ± 0.35	54.55 ± 0.38	+0.32

Table 12: Drop-one-out ablation on CIFAR-100 (full data, 5 seeds). Every reverted mechanism lowers accuracy; the pure baseline is -1.26 below full SD-TTC.

Configuration	Acc. (%)	Δ Acc.
Full SD-TTC	54.55 ± 0.38	—
w/o Fiedler gate (fixed $\alpha=0.25$)	54.06 ± 0.24	−0.49
w/o Toda frame (static FOP)	53.94 ± 0.28	−0.61
Pure baseline (all reverted)	53.29 ± 1.02	−1.26

The standard probe matches or exceeds the energy read-out on every dataset, confirming that SD-TTC learns linearly-separable representations.

A.7 DROP-ONE-OUT ABLATION ON CIFAR-100

Table 12 repeats the drop-one-out ablation on the harder CIFAR-100 (5 seeds). In contrast to CIFAR-10—where the small $+0.17/ +0.23$ deltas fall within run-to-run noise (§4.8)—every mechanism contributes *positively* here, and the full model is best, confirming that the components act synergistically once the task is hard enough to expose them.

A.8 HEAD-TO-HEAD EFFICIENCY FRONTIER (VISUALIZATION)

Fig. 9 visualizes the controlled head-to-head of Table 4: SD-TTC concedes a small accuracy margin to SCFF and block-wise SSL but consistently dominates on the efficiency axis.

A.9 LOCAL ANTI-COLLAPSE EXTENSIONS ON TINY-IMAGENET

Table 13 details the three strictly-local, single-pass extensions to $\mathcal{L}^{(l)}$ introduced in §4.2, evaluated with a block-wise greedy ResNet-18 on Tiny-ImageNet (200 classes) via a linear probe on the frozen, concatenated Phase-1 features. **MSP** pools feature maps at scales $\{1 \times 1, 2 \times 2, 4 \times 4\}$ before density projection; **VR** applies a variance hinge $\max(0, 1 - \text{std}(Z))$ on the projected read-out Z to prevent dimensional collapse; **APH** adds a local squeeze-and-excite gate $1 + 0.5 \tanh(\cdot)$ prior to the Toda frame. All operate *within* each block during a single forward pass—no global backprop or top-down feedback. Local attention (APH) gives the largest gain ($+1.12$), while VR most effectively restores mid-stage effective dimensionality (ED), and throughput stays at 8.8–9.7k img/s throughout (Fig. 10).

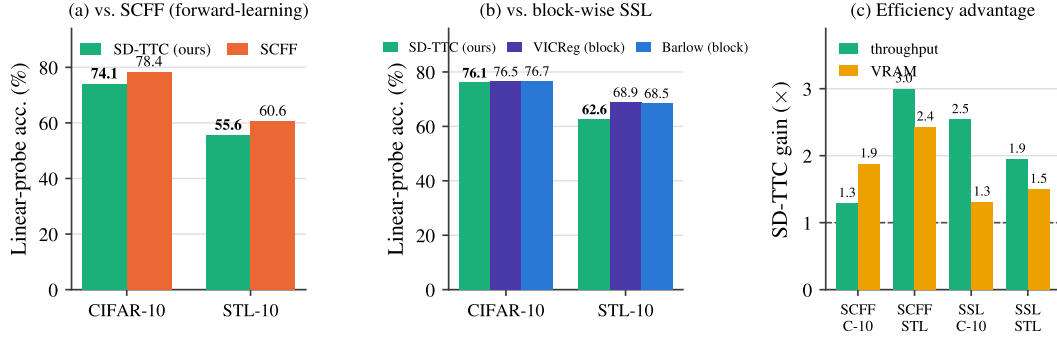


Figure 9: Head-to-head under an identical 3-block CNN. (a) SD-TTC vs. SCFF and (b) vs. greedy block-wise VICReg/Barlow-Twins linear-probe accuracy; (c) SD-TTC’s efficiency advantage (throughput speed-up and peak-VRAM reduction) over each baseline. SD-TTC trades a small accuracy margin for consistently 1.3–3.0 $\times$ better efficiency.

Table 13: Local extensions on Tiny-ImageNet (block-wise ResNet-18, linear probe on frozen concatenated Phase-1 features). ED columns are per-stage effective dimensionality; throughput is Phase-1 img/s. Best per column in **bold**.

Configuration	MSP	VR	APH	Concat probe (%)	Final-stage (%)	Throughput (img/s)	Stage-1 ED	Stage-2 ED
Base SD-TTC				15.17	11.60	9732	29.93	31.90
+ MSP	✓			15.44 +0.27	12.05	9332	30.53	32.42
+ VR		✓		15.72 +0.55	11.98	9281	34.24	34.21
+ APH			✓	16.29 +1.12	12.52	9038	36.09	34.98
Full (MSP+VR+APH)	✓	✓	✓	16.17 +1.00	12.48	8791	41.52	15.47

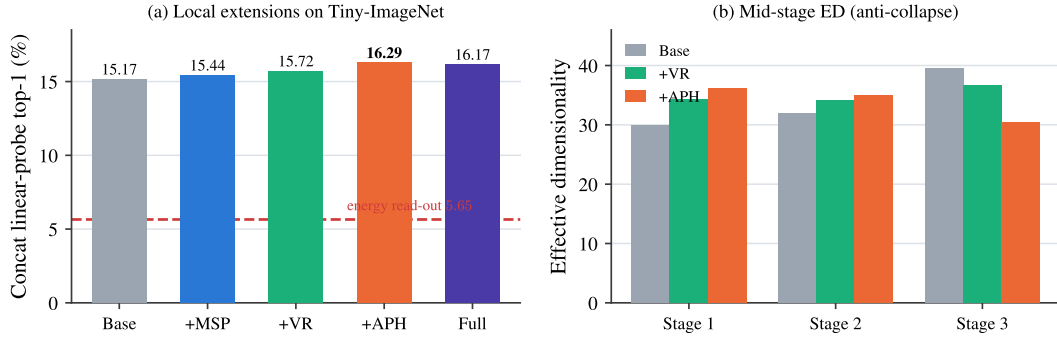


Figure 10: Local anti-collapse extensions on Tiny-ImageNet. (a) Concatenated linear-probe top-1 for each local extension; all far exceed the uncalibrated $\mathbb{E}[\hat{Y}^2]$ energy read-out (dashed). (b) Per-stage effective dimensionality: the variance hinge (VR) and local attention (APH) lift mid-stage ED, counteracting hierarchical abstraction collapse.

A.10 EXTENDED 10-SEED ABLATIONS: SHEAF DIFFUSION AND THE FIEDLER GATE

To settle whether Sheaf diffusion (A) and the Fiedler gate (C) contribute beyond run-to-run noise (§4.8), we run a 10-seed study on CIFAR-10 and CIFAR-100. Table 14 reverts Sheaf diffusion to $N = 20$ multi-pass input dropout: the two are *statistically indistinguishable* in accuracy ($p > 0.15$), confirming that the sheaf mechanism’s value is *efficiency*—a $\sim 4\times$ Phase-1 speed-up and a $7.5\times$ peak-memory reduction—rather than an accuracy change. Table 15 contrasts the adaptive gate with the best fixed constants: the optimal constant *flips* across datasets ($\alpha = 0.10$ on CIFAR-10, $\alpha = 0.25$ on CIFAR-100), and mis-setting it costs up to -0.83 on CIFAR-100, whereas the single adaptive gate matches or exceeds the tuned constant on both datasets with no per-dataset search.

Table 14: Sheaf diffusion vs. $N=20$ multi-pass input dropout (10 seeds). Accuracy is statistically tied (p from a paired t -test); the sheaf mechanism buys a large Phase-1 speed-up and memory reduction.

Dataset	Full SD-TTC (Sheaf)	Reverted $N=20$	ΔAcc (p)	Ph-1 (s/ep)	Reverted (s/ep)	Speed-up	Peak VRAM
CIFAR-10	77.41 ± 0.26	77.60 ± 0.28	-0.18 (0.151)	6.1	23.9	$3.95\times$	1853 / 13840 MB
CIFAR-100	54.48 ± 0.43	54.63 ± 0.38	-0.15 (0.412)	6.4	24.0	$3.74\times$	1854 / 13841 MB

Table 15: Fiedler gate vs. fixed compression constants (10 seeds). The best fixed constant flips across datasets; the adaptive gate is best-or-tied on both. Best per row in **bold**.

Dataset	Fixed $\alpha=0.10$	Fixed $\alpha=0.25$	Adaptive Fiedler ($\gamma=0.5$)	Optimal constant
CIFAR-10	77.59 ± 0.37	77.52 ± 0.38	77.41 ± 0.26	$\alpha=0.10$
CIFAR-100	53.60 ± 0.37	54.42 ± 0.30	54.48 ± 0.43	$\alpha=0.25$